SMART INDIA HACKATHON 2026

AI-Powered Risk & Relocation Decision Support System

Problem Statement ID: 26191

Intelligent Identification of Hazard-Based Red Zones, Carrying Capacity Assessment, and Immediate Relocation Needs for Vulnerable Habitations

Theme: Disaster Management

PS Category: Software

Team ID: 165338

Team Name: Apexx_

Department: Department of Computer Engineering

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Team Members

1. **Prathamesh Shelke** (Team Leader)
2. **Krish Rai**
3. **Anamika Jha**
4. **Sarang Zade**
5. **Omkar Tandale**
6. **Jaykumar Yadav**

2. ABSTRACT

Many rural settlements across India are located in areas that face repeated landslides, floods, and ground sinking during the monsoon season. Right now, disaster management mostly focuses on rescuing people and setting up temporary relief camps after a disaster happens. Moving vulnerable communities before a disaster strikes requires identifying high-risk zones, finding safe nearby land, and checking if that new land has enough water, road access, and space to support the population. Currently, doing this through manual surveys and separate departmental files takes around 3 months.

Our goal is to build an **AI-Powered Risk & Relocation Decision Support System** that helps authorities identify hazardous habitations, assess carrying capacity, and prioritize vulnerable communities for timely relocation. The system follows a simple **Detect → Assess → Act** workflow. It uses **ISRO Bhuvan 10m DEM** and **Sentinel-1 & 2** satellite data on a 10m × 10m grid (which is 9 times sharper than standard 30m maps) along with a hybrid **XGBoost + AHP** model (60% AI + 40% terrain rules) to predict a habitation risk score (0–100%). It then searches within a 5–25 km radius for safe host sites with slope under 12°, checks 5+ carrying capacity parameters based on **NDMA norms** (3.5 sq.m land and 55L water per person per day), and generates **Immediate, High, Medium, and Low** relocation priority lists, cutting planning time from 3 months down to 48 hours.

3. INTRODUCTION

Current Situation

Thousands of villages in hilly regions and floodplains face heavy damage every monsoon. While emergency teams work hard to evacuate people during heavy rains, families often return to the same unsafe homes afterward because permanent, verified relocation sites are not ready.

Why the Problem Exists

Terrain stability changes over time due to heavy rain, deforestation, and slope cutting. However, district offices still use old static maps and outdated population census numbers. Also, satellite data, weather records, and infrastructure maps are kept in separate departments rather than in one connected tool.

Who Is Affected

Villagers living in high-risk settlements risk losing their lives, homes, and farms. At the same time, State and District Disaster Management Authorities (SDMAs, DDMAs) and District Magistrates (Collectors) struggle to quickly decide which villages must be moved first and where they can settle safely without causing overcrowding.

Why Existing Methods Are Not Enough

Current processes rely on manual field analysis and static maps that do not reflect real-time terrain or weather changes. There is no automated system to rank relocation priority or verify whether a proposed resettlement plot has sufficient water, road access, schools, and hospitals.

4. PROBLEM STATEMENT

Problem Statement ID **26191** (Theme: Disaster Management, Category: Software) is titled **"Intelligent Identification of Hazard-Based Red Zones, Carrying Capacity Assessment, and Immediate Relocation Needs for Vulnerable Habitations."**

The problem statement requires a software solution that addresses three core tasks:

1. **Dynamic Red Zone Identification:** Automatically detect and map high-hazard zones (Red Zones) using terrain, satellite, and multi-hazard data rather than relying on static maps.
2. **Carrying Capacity Assessment:** Identify safe nearby host sites and calculate how many people they can sustainably support based on land area, slope, water supply, healthcare, schools, and road access.
3. **Relocation Prioritization:** Rank vulnerable settlements into clear priority tiers so authorities can plan immediate and phased pre-monsoon relocations.

5. OBJECTIVES

1. **Predict Habitation Risk Using AI:** Use an XGBoost model combined with AHP terrain rules (60% AI + 40% rules) to compute a 0–100% risk score for each settlement with a target accuracy above 90%.
2. **Map Sharp 10m × 10m Risk Grids:** Use ISRO Bhuvan 10m DEM and Sentinel-1/2 data with QGIS/Leaflet to map 4 risk zones and highlight Red Zones (score ≥ 0.8).
3. **Locate Safe Host Sites (5–25 km):** Automatically filter safe areas within 5 to 25 km of vulnerable villages where the terrain slope is less than 12° so villagers stay close to their farms and jobs.
4. **Evaluate 5+ Carrying Capacity Parameters:** Check population, water, healthcare, roads, and schools using NDMA relief norms (3.5 sq.m land and 55L water/day per person).
5. **Automate Relocation Priority Lists:** Classify habitations into **Immediate (<30 days), High, Medium (1–6 months), and Low** relocation priorities and support single or split-site matching.
6. **Support Fast & Offline Field Operations:** Deliver results in under 45 seconds (target response <5 minutes) and provide an offline Progressive Web App (PWA) using lightweight PMTiles maps (<50 MB).

6. EXISTING SYSTEM

Currently, identifying unsafe villages and finding land for resettlement is handled through manual paperwork and separate departmental surveys:

- **Manual Analysis:** Field teams inspect villages one by one after complaints of land cracks or flooding are reported, taking around 3 months to prepare a relocation plan.
- **Static Maps:** Officials use coarse, older PDF maps that cannot be updated quickly when heavy rainfall occurs.
- **Separate Data Sources:** Elevation maps, rainfall records, census counts, and road/hospital locations are stored in different formats across different offices.
- **No Automated Relocation Priority or Capacity Check:** Resettlement plots are often chosen based only on vacant land availability without calculating whether local water sources, roads, and health centers can handle the extra population.

7. GAPS IN EXISTING SYSTEM

Existing Approach	Limitation	Our Improvement
Manual analysis	Slow inspection process that takes up to 3 months for planning	**AI-Based Risk Prediction** (XGBoost + AHP cuts planning time to 48 hours)
Static Map	Low resolution and does not update with new weather or terrain data	**Dynamic GIS Mapping** (10m × 10m grid maps via QGIS/Leaflet, 9x sharper)
Separate Data	Terrain, weather, population, and infrastructure data are disconnected	**Satellite + DEM + AI Integration** (Sentinel-1/2, Bhuvan 10m DEM, IMD, OSM)
No Relocation Priority	No systematic ranking of which village must move first or where	**Automated Relocation Planning** (Immediate / High / Medium / Low priority + capacity fit)

8. PROPOSED SOLUTION

Our proposed system is an **AI-Powered Risk & Relocation Decision Support System** built around three clear steps: **Detect** → **Assess** → **Act**.

- **Step 1 – Detect (AI + Satellite/DEM Data):** The system ingests **ISRO Bhuvan 10m DEM** slope maps, **Sentinel-1** all-weather radar (which works even through heavy monsoon clouds), **Sentinel-2** optical imagery, and **IMD rainfall API** feeds. It divides the terrain into a 10m × 10m grid and runs our hybrid **XGBoost + AHP model** to calculate a habitation risk score from 0 to 100%. Areas with a score ≥ 0.8 are marked as **Red Zones**.
- **Step 2 – Assess (Population + Infrastructure Capacity):** Since census data can be outdated, the system uses **AI rooftop counting** along with census records to estimate current village population. It then scans safe zones within a **5–25 km radius** with gentle slopes (**<12°**) and checks 5+ infrastructure parameters (water, roads, hospitals, schools, land). Using **NDMA relief norms** (3.5 sq.m land and 55L water/day per person), it computes net available capacity (`Safe Capacity - Existing Population`) and determines whether a **Single Site** or a **Split Site** (dividing households between two nearby sites) is needed.
- **Step 3 – Act (Relocation Priority + GIS Alerts):** Each habitation is assigned an **Immediate (<30 days), High, Medium (1–6 months), or Low** priority. **SHAP explainability** scores show the District Collector the exact factors behind the AI prediction. Officials can view the interactive React + Leaflet map, work offline via PWA (<50 MB PMTiles), and export 1-click PDF or GeoJSON reports for final sign-off.

9. SYSTEM WORKFLOW

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1. GIS Data Preparation (ISRO Bhuvan 10m DEM + Sentinel-1/2 Radar +
Village/Rooftop Population + OSM Rivers, Roads, Hospitals, Schools)
↓ 2. AI Risk Assessment (XGBoost + 40% AHP Terrain Rules →
Calculates Hazard + Vulnerability Risk Score [0–100%] + SHAP
Explainability) ↓ 3. Safe Site Identification (Searches Safe Zones
within 5–25 km radius → Filters slope < 12° → Checks
road, water, health & school access) ↓ 4. Carrying Capacity
Assessment (Applies NDMA Norms: 3.5 sq.m land & 55L water/day per
person → Available Capacity [Safe - Existing] → Single or
Split Site fit) ↓ 5. Relocation Priority & Authority Dashboard
(Generates Immediate / High / Medium / Low Priority List → React +
Leaflet Interactive Map → 1-Click PDF/GeoJSON Export → DM
Action)
```

10. SYSTEM ARCHITECTURE

The system architecture is organized into four practical layers matching our technical implementation:

1. **Data Inputs Layer:** Collects Satellite/DEM imagery (Sentinel-1 & 2, ISRO Bhuvan 10m DEM), Census & AI rooftop population counts, OpenStreetMap (OSM) infrastructure networks, and IMD weather/hazard records.
2. **Processing & Analysis Layer (GIS + AI):** Uses **QGIS / GDAL** and **PostGIS** for spatial grid alignment (10m × 10m), runs the **XGBoost + AHP** model to map the 4 risk zones (Red Zone ≥ 0.8), performs the 5–25 km safe-site spatial search, and runs the carrying capacity verification.
3. **Backend Layer:** Built on **Python FastAPI**, which connects the spatial database (PostGIS), AI analysis outputs, and OGC-compliant REST map services to generate four core outputs: Red Zone Map, Ranked Safe Sites, Fit Verification, and the Authority Decision Tool.
4. **Authority Dashboard Layer (User Layer):** A **React + Leaflet** web application with **PWA offline mode** and **PMTiles (<50 MB)** support. It displays the risk map, relocation priority list, SHAP explanations, and provides 1-click PDF and GeoJSON exports.

11. TECHNOLOGY STACK

Technology	Purpose
React.js + Leaflet.js	Frontend interactive GIS map dashboard, layer controls, and priority list tables
PWA + PMTiles (<50 MB)	Progressive Web App for offline field access and lightweight vector map tiles
Python + FastAPI	High-speed backend connecting PostGIS data, ML models, and OGC REST APIs (<45s processing)
PostgreSQL + PostGIS	Spatial database storing 10m × 10m grids, habitation polygons, and running 5–25 km radius queries
XGBoost + AHP + SHAP	Hybrid AI risk prediction (60% XGBoost + 40% AHP terrain rules) with SHAP explainability scores
QGIS	GIS terrain preprocessing, slope calculation, and 4-zone hazard mapping
ISRO Bhuvan 10m DEM & Sentinel-1/2	High-resolution 3D slope maps (10m) and all-weather cloud-penetrating radar satellite data
IMD & OpenStreetMap (OSM) APIs	Free rainfall data feeds and road, river, school, hospital, and building rooftop data

12. AI/ML COMPONENT

- **Data Used:** ISRO Bhuvan 10m DEM (slope, elevation, aspect), Sentinel-1 SAR radar (all-weather terrain deformation and moisture), Sentinel-2 land cover, IMD rainfall data, historical disaster records, and AI rooftop counts from satellite imagery.
- **Preprocessing:** Terrain and environmental layers are resampled into uniform **10m × 10m grid cells** (providing 9x sharper resolution than standard 30m grids). Road, river, school, and hospital distances are calculated using PostGIS spatial functions.
- **Hybrid Model (60% XGBoost + 40% AHP):** Relying purely on machine learning fails in remote villages that have very few past recorded disaster entries. To solve this, we combine **60% XGBoost** probability scores with **40% Analytic Hierarchy Process (AHP)** domain rules based on physical slope and drainage proximity.
- **Prediction & Classification:** The model outputs a 0–100% Risk Score for each habitation and classifies areas into 4 risk zones (with score ≥ 0.8 marked as a Red Zone) and 4 relocation priority tiers (Immediate, High, Medium, Low). Based on reference benchmarks, the XGBoost + AHP approach achieves **91.4% accuracy** (target >90%).
- **Explainability with SHAP:** Every prediction is paired with a **SHAP (SHapley Additive exPlanations)** score breakdown so the District Collector can see exactly how much slope, rainfall, or river proximity contributed to a village's risk rating.

13. DATABASE DESIGN

Table Name	Important Fields	Purpose
`risk_grid_10m`	`grid_id` (PK), `geom` (Polygon), `slope_deg`, `xgb_prob`, `ahp_score`, `final_risk`, `zone_class`	Stores the 10m × 10m spatial grid cells and identifies Red Zones (≥0.8)
`habitations`	`hab_id` (PK), `village_name`, `geom`, `rooftop_count`, `est_population`, `risk_score`, `shap_json`, `priority_tier`	Stores vulnerable settlement data, AI rooftop population estimates, SHAP reasons, and relocation priority
`safe_host_sites`	`site_id` (PK), `geom`, `slope_deg` (<12°), `land_sqm`, `water_lpd`, `road_dist`, `health_dist`, `school_dist`, `avail_capacity`	Stores candidate safe sites within 5–25 km and their NDMA-verified net carrying capacity
`relocation_plans`	`plan_id` (PK), `hab_id` (FK), `site_id` (FK), `allocation_type` (Single/Split), `moved_pop`, `dm_signoff`	Stores the matched village-to-host relocation plan and District Magistrate approval status

14. MODULES OF THE SYSTEM

1. **GIS Data Preparation Module:** Fetches Bhuvan 10m DEM, Sentinel-1/2 data, IMD rainfall feeds, and OSM infrastructure layers (rivers, roads, schools, hospitals) and prepares the 10m × 10m spatial database in PostGIS.
2. **AI Risk Assessment Module:** Runs the 60% XGBoost + 40% AHP hybrid engine to compute hazard and vulnerability scores (0–100%), delineates Red Zones (≥0.8), and calculates SHAP explanations.
3. **Safe Site Identification Module:** Searches within a 5–25 km radius around vulnerable habitations, filters for safe terrain with slope < 12°, and checks road, water, healthcare, and school proximity.
4. **Carrying Capacity Assessment Module:** Applies NDMA relief rules (3.5 sq.m land and 55L water/day per person), subtracts existing occupancy from total safe capacity, and generates Single or Split Site allocation options.
5. **Relocation Priority & Authority Dashboard Module:** Ranks habitations into Immediate (<30 days), High, Medium (1–6 months), and Low queues, renders the interactive React + Leaflet map, supports offline PWA usage (<50 MB PMTiles), and exports PDF/GeoJSON files.

15. USER INTERFACE

- **Login & Jurisdiction Screen:** Secure login for State (SDMA) and District (DDMA / District Collector) officials.
- **Main GIS Risk Map Screen:** React + Leaflet interactive map displaying the 4 risk zones on a 10m × 10m grid, highlighting Red Zones (≥0.8) and vulnerable village markers.
- **Habitation SHAP & Risk Analysis Screen:** Displays a habitation's risk score (0–100%), estimated population from AI rooftop counting, and a visual bar chart of SHAP feature contributions.
- **Safe Site & Carrying Capacity Screen:** Shows ranked safe host sites within 5–25 km (slope < 12°), detailing their available capacity across 5+ parameters (land, water, roads, healthcare, schools) and Single/Split site fit verification.

- **Relocation Priority & Export Panel:** Lists villages grouped under Immediate (<30 days), High, Medium (1–6 months), and Low priority, with 1-click PDF and GeoJSON export buttons and a District Magistrate sign-off toggle.

16. FEASIBILITY ANALYSIS

A. Technical Feasibility

- **Proven Stack:** Built using PostGIS, FastAPI, XGBoost, and AHP, achieving 91.4% benchmark accuracy and processing updates in under 45 seconds.
- **All-Weather Data Availability:** Uses free Sentinel-1 Radar data (which penetrates clouds during monsoons) alongside ISRO Bhuvan 10m DEM, IMD rainfall APIs, and OpenStreetMap.
- **Fast & Offline Ready:** Uses PMTiles to compress district maps under 50 MB for offline PWA use.

B. Operational Feasibility

- **Authority-Driven & Policy Compliant:** Transparent SHAP explanations help officials verify recommendations, while final relocation orders require the District Magistrate's (DM) sign-off in compliance with NDMA guidelines.
- **Easy State/District Integration:** Connects with 36 SDMAs and 700+ DDMAs across India using standard OGC REST APIs.

C. Economic & Financial Feasibility

- **Zero License Cost (80% Cheaper):** Built on a 100% open-source software stack, making it roughly 80% cheaper than proprietary commercial GIS tools like ArcGIS.
- **₹0 Data Cost:** Relies on free government and open APIs (ISRO Bhuvan, IMD, Copernicus, OSM).
- **High ROI:** Proactive pre-monsoon relocation reduces post-disaster emergency relief spending by an estimated 35–40%.

17. CHALLENGES AND RISKS

1. **Heavy Monsoon Clouds & Outdated Population Records:** Optical satellites cannot see through dense monsoon clouds, and 2011 census figures do not reflect current village populations.
2. **Limited Past Disaster Data:** Some remote villages lack digital historical records of past landslides or flash floods for training purely data-driven ML models.
3. **No Internet Connectivity in Remote Areas:** Field officers inspecting hilly or rural host sites often have poor or no mobile network coverage.
4. **False Alerts & Villager Pushback:** Inaccurate risk alerts can cause panic, and villagers often resist moving if the new host site is far from their agricultural fields.

18. MITIGATION STRATEGIES

Challenge / Risk	Possible Impact	Mitigation (How We Solve It)
Clouds & Old Population Data	Blocked satellite view during rain and inaccurate headcount for capacity planning	Use **Sentinel-1 Radar satellites** (works through heavy clouds) + **AI rooftop counting** to estimate current population
Less Past Disaster Data	ML model may underperform in unrecorded areas	Combine **60% AI + 40% AHP slope/terrain rules** so the system works reliably even without past records
No Internet in Remote Areas	Field teams cannot load maps on site	Provide **PWA offline mode** with lightweight **PMTiles maps (under 50 MB)** and **1-click PDF downloads**
False Alerts & Villager Pushback	Community refusal to relocate to distant plots	Keep host site search within **5–25 km** (near farms/jobs) + require **final approval by the District Collector (DM)**

19. UNIQUENESS OF OUR SOLUTION

- **Manual Analysis → AI-Based Risk Prediction:** Replaces 3-month manual screening with an XGBoost + AHP engine that prepares plans in 48 hours.
- **Static Map → Dynamic GIS Mapping:** Upgrades coarse static maps to dynamic 10m × 10m grid maps (9x sharper) updated within 5 minutes of new data.
- **Separate Data → Satellite + DEM + AI Integration:** Unifies Sentinel-1/2 radar, Bhuvan 10m DEM, IMD rain feeds, and OSM infrastructure into one PostGIS workflow.
- **No Relocation Priority → Automated Relocation Planning:** Automatically generates Immediate, High, Medium, and Low priority queues and verifies Single or Split host site capacity using NDMA norms.

20. INNOVATION

- **Hybrid 60% AI + 40% AHP Risk Engine with SHAP:** Balances machine learning precision with physical terrain rules and explains every risk score clearly to officials.
- **Cloud-Penetrating Radar + AI Rooftop Triage:** Uses Sentinel-1 SAR data for all-weather monitoring and counts building rooftops from imagery to overcome outdated census data.
- **NDMA-Compliant Single & Split Host Site Matching:** Automatically calculates net carrying capacity (3.5 sq.m land and 55L water/day per person) within a livelihood-friendly 5–25 km radius and splits allocations across multiple safe sites when a single plot is too small.
- **Sub-50 MB Offline PWA Architecture:** Uses PMTiles so field teams can inspect high-resolution 10m risk grids on mobile or laptop browsers without internet connectivity.

21. SECURITY AND PRIVACY

- **Authentication & Role-Based Authorization:** Secure login for SDMA, DDMA, and District Collectors. Only authorized District Magistrates can mark a relocation plan as approved.
- **User Privacy Protection:** The platform processes settlement-level rooftop counts and aggregate population numbers; no personal citizen identifiers (such as Aadhaar or bank details) are stored.
- **Secure API & Database Standards:** FastAPI endpoints use HTTPS and input validation, while spatial records are protected inside PostgreSQL/PostGIS with OGC-compliant access controls.

22. IMPLEMENTATION PLAN

- **Stage 1 – Requirement Analysis:** Define multi-hazard parameters, NDMA relief norms (3.5 sq.m land, 55L water/day), and SDMA/DDMA workflow needs.
- **Stage 2 – GIS Data Preparation:** Collect ISRO Bhuvan 10m DEM, Sentinel-1/2 imagery, IMD rainfall data, and OpenStreetMap layers (roads, rivers, schools, hospitals) and generate 10m × 10m grids.
- **Stage 3 – AI/ML Development:** Train the XGBoost model, configure the 40% AHP terrain rule weights, and integrate SHAP explainability.
- **Stage 4 – Backend Development:** Set up PostGIS spatial queries (5–25 km safe site search, slope < 12° filter, Single/Split capacity math) and build FastAPI REST endpoints.
- **Stage 5 – UI & Offline PWA Development:** Build the React + Leaflet dashboard, integrate PMTiles (<50 MB) offline caching, and add 1-click PDF/GeoJSON export.
- **Stage 6 – Integration:** Connect the FastAPI backend, PostGIS database, and React frontend to achieve sub-45-second end-to-end processing.
- **Stage 7 – Testing:** Validate spatial queries, carrying capacity calculations, offline mode, and model accuracy.
- **Stage 8 – Deployment:** Package the application for state and district deployment via OGC REST APIs.

23. TESTING

Testing covers functional verification of the NDMA capacity math, PostGIS spatial query testing, FastAPI endpoint testing, XGBoost + AHP validation, and offline PWA load testing.

Test Case	Input	Expected Output	Result
TC-01: Red Zone Detection	10m × 10m grid cell with steep slope, heavy IMD rain, and high radar moisture	XGBoost + AHP outputs risk score ≥ 0.8, marks Red Zone, and generates SHAP breakdown	To be tested
TC-02: Safe Host Site Filter	Query candidate host sites around a Red Zone village	Returns only safe sites within 5–25 km radius having slope < 12° and road/water/school/health access	To be tested
TC-03: NDMA Carrying Capacity Math	Host site available land area and daily water supply	Calculates net capacity (`Safe - Existing`) enforcing ≥ 3.5 sq.m land and ≥ 55L water/day per person	To be tested
TC-04: Split Site Allocation	Habitation population exceeds capacity of the nearest single safe site	System automatically generates a Split Site plan across two nearby safe sites within 5–25 km	To be tested
TC-05: Speed & Offline PWA Test	Trigger data update and switch browser to offline mode	Update finishes in <45s; offline PWA renders PMTiles map (<50 MB) and exports PDF report	To be tested

24. EXPECTED RESULTS

- **Faster Relocation Planning (3 Months → 48 Hours):** Automates data fusion and site screening so authorities can finalize pre-monsoon relocation lists in 48 hours.
- **9x Sharper Risk Mapping:** 10m × 10m grid resolution identifies localized slope and flood risks with >90% target accuracy (91.4% benchmark).
- **Actionable Triage Queues:** Clear categorization of habitations into Immediate (<30 days), High, Medium (1–6 months), and Low relocation timelines.
- **Sustainable Resettlement:** Verified host sites that meet NDMA land and water norms without overcrowding existing communities.

25. SOCIAL AND PRACTICAL IMPACT

1. **Social Impact (For Villagers):** Enables proactive pre-monsoon family relocation, preserves livelihoods by keeping host sites within a 5–25 km radius (near farms and jobs), and guarantees NDMA relief norms (3.5 sq.m land and 55L water/day per person).
2. **Operational Benefits (For Authorities):** Cuts planning time from 3 months to 48 hours, provides pinpoint 10m × 10m maps, and delivers transparent SHAP scores with 1-click PDF/GeoJSON outputs for the District Collector.
3. **Economic Benefits (For Government):** Reduces post-disaster emergency relief spend by 35–40%, lowers software technology costs by 80% using a 100% open-source stack (no ArcGIS license), and incurs ₹0 data cost via free APIs.
4. **Long-Term & Environmental Impact:** Prevents host-site overcrowding, enforces a Red-Zone freeze to block risky new construction, and aligns with NDMA and the UN Sendai Framework.

26. FUTURE SCOPE

- **Nationwide Rollout via OGC APIs:** Plug the platform directly into all 36 SDMAs and 700+ DDMAs across India using standardized OGC REST APIs.
- **IoT Ground Sensor Integration:** Connect real-time rain gauges and slope inclinometers in Immediate-priority villages to trigger automated threshold alerts.
- **Direct Land Record (Bhulekh) Integration:** Link state land ownership APIs to automatically separate vacant government land from private land during host site screening.
- **Multilingual Field Reports:** Generate exported PDF relocation summaries in regional Indian languages for local panchayat and block officers.

27. LIMITATIONS

- **Mandatory Ground Verification:** While the platform narrows down safe host sites and calculates capacity, physical soil load testing and final sign-off by the District Magistrate remain necessary before construction.
- **Internet Needed for Live Satellite Sync:** Although the PWA and PMTiles maps work completely offline in the field, syncing new IMD rainfall feeds or Sentinel imagery requires internet connectivity at the district office.
- **Estimated Household Size:** AI rooftop counting accurately detects building footprints, but total population is estimated by multiplying rooftop counts with regional average household sizes.

28. CONCLUSION

Shifting disaster management from post-disaster rescue to pre-monsoon relocation requires fast, accurate, and practical planning tools. Addressing Problem Statement 26191, our **AI-Powered Risk & Relocation Decision Support System** uses a clear **Detect** → **Assess** → **Act** workflow to map hazardous Red Zones, evaluate host site carrying capacity, and prioritize vulnerable habitations.

By combining **ISRO Bhuvan 10m DEM**, **Sentinel-1/2 all-weather radar**, **PostGIS**, and a hybrid **XGBoost + AHP (60% AI + 40% terrain rules)** model with **SHAP explainability**, the system produces 9x sharper risk maps on a 10m × 10m grid. It searches within a livelihood-friendly 5–25 km radius for safe sites (slope < 12°) and verifies 5+ capacity parameters against **NDMA norms** (3.5 sq.m land and 55L water/day per person). Built on a 100% open-source stack with offline PWA support, our solution cuts technology costs by 80%, reduces emergency relief spending by 35–40%, and shortens relocation planning from 3 months to 48 hours.

29. REFERENCES

1. **ISRO Bhuvan (`bhuvan.nrsc.gov.in`):** 10m Digital Elevation Model (DEM) slope maps and 10m × 10m spatial grid processing.
2. **Copernicus Sentinel-1 & 2 (`dataspace.copernicus.eu`):** All-weather Synthetic Aperture Radar (works through heavy clouds) and optical terrain risk data.
3. **National Disaster Management Authority – NDMA Norms (`ndma.gov.in`):** Minimum relief and resettlement norms of 3.5 sq.m land and 55 liters of water per person per day.

4. **India Meteorological Department & OpenStreetMap (`mausam.imd.gov.in`, `openstreetmap.org`):** Free rainfall APIs, road networks, schools, healthcare facilities, and building footprints for 5+ capacity checks within 5–25 km.
5. **UNDRR Sendai Framework & IEEE Geo-AI Literature (`ieeexplore.ieee.org`, `undrr.org`):** Benchmarks for hybrid XGBoost + AHP susceptibility modeling (91.4% accuracy), rapid disaster triage, and proactive risk reduction.
6. **Smart India Hackathon 2026 Portal (`sih.gov.in`):** Problem Statement ID 26191 official problem description.